

Sri Lanka Institute of Information Technology

Programing Paradigms SE2062

EBNF of DSL and HTML Form Generator with Lex and Yacc

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```

program      ::= "form" IDENTIFIER "{" form_body validate_block?"}"
form_body    ::= meta_section? section_list
meta_section ::= { "meta" IDENTIFIER "=" STRING ";"
}
section_list ::= { "section" IDENTIFIER "{" field_list "}"
}
field_list   ::= { field
}

field        ::= "field" IDENTIFIER ":" field_type attribute_block?
";"
              | "field" IDENTIFIER ":" field_type string_list
attribute_block? ";"
field_type    ::= "text" | "password" | "number" | "email" | "date"
              | "checkbox" | "radio" | "dropdown" | "textarea" |
"file" string_list ::= "[" STRING { "," STRING } "]"
attribute_block ::= { attribute
}
attribute     ::= "required"
              | "min" "=" (NUMBER | STRING)
              | "max" "=" (NUMBER | STRING)
              | "default" "=" (STRING | BOOL)
              | IDENTIFIER "=" (NUMBER | STRING)
validate_block ::= "validate" "{" { validation_rule } "}"
validation_rule ::= "if" condition "{" "error" STRING ";" "}"
condition       ::= IDENTIFIER comparator NUMBER
comparator       ::= "<" | ">" | "=="

```

STRING ::= '"' (any character except quote)+ '"'
IDENTIFIER ::= (letter)(letter | digit | "_")*
NUMBER ::= digit+
BOOL ::= "true" | "false"

❖ Program Structure

```
program ::= "form" IDENTIFIER "{" form_body validate_block? "}"
```

- Defines the entire form.
- Starts with form, a name (IDENTIFIER), followed by the form body.
- Optionally ends with a validation block.

❖ Form Body

```
form_body ::= meta_section? section_list
```

- Contains optional metadata and one or more sections.

❖ Metadata

```
meta_section ::= { "meta" IDENTIFIER "=" STRING ";" }
```

- Allows adding form-level metadata like author or version.

❖ Sections

```
section_list ::= { "section" IDENTIFIER "{" field_list "}" }
```

- Defines named sections within the form.
- Each section contains a list of fields.

❖ Fields

```
field_list ::= { field }  
field ::= "field" IDENTIFIER ":" field_type attribute_block?  
";"  
      | "field" IDENTIFIER ":" field_type string_list  
attribute_block? ";"
```

- Fields are form inputs (text, number, etc.).
- Some field types (like dropdown, radio) include option lists.



❖ Field Types

```
field_type ::= "text" | "password" | "number" | ...
```

- Specifies the type of input (e.g., email, checkbox, file, etc.).

❖ Option Lists

```
string_list ::= "[" STRING { "," STRING } "]"
```

- Used in fields like radio and dropdown.
- Specifies selectable options.

❖ Attributes

```
attribute_block ::= { attribute }  
attribute ::= "required" | "min=..." | "default=..." | ...
```

- Additional settings for fields like required, min, max, etc.

❖ Validation Rules

```
validate_block ::= "validate" "{" { validation_rule } "}"
```

- Optional block to define logic checks on input.

❖ Conditions

```
validation_rule ::= "if" condition "{" "error" STRING ";" "}"  
condition ::= IDENTIFIER comparator  
NUMBER  
comparator ::= "<" | ">" |  
"=="
```

- Simple validation checks like if age < 18, showing an error if true.

❖ Terminals

```
STRING ::= "\"" text "\""   
IDENTIFIER ::= letter followed by letters/digits/underscores  
NUMBER ::= digits  
BOOL ::= "true" | "false"
```

- These are the basic units of the language: strings, IDs, numbers, booleans.

Terminals

Terminal Type	Examples
Keywords	form, meta, section, field, validate, if, error, required, true, false
Symbols	{, }, [,], :, ;, =, <, >, ==, ,
Tokens	STRING, IDENTIFIER, NUMBER, BOOL

Non-terminals

program
form_body
meta_section
section_list
field_list field
field_type
string_list
attribute_block
attribute
validate_block
validation_rule
condition comparator